

Q I've been offered a chorionic villus sampling (CVS) test. What's a chorionic villus?

The term chorionic refers to your placenta, which is covered in tiny fingerlike fronds known as villi (singular, villus). These fronds increase the surface area of your placenta, maximizing the transfer of nutrients to your baby's body and waste out of it. The villi contain genetic material that is identical to your baby's, so the CVS test harvests some villi to analyze your baby's genetic makeup.

Q What are the risks associated with invasive tests?

Any invasive procedure carries with it a risk of infection, so be aware of any swelling, heat, or redness at the puncture site on your abdomen, if relevant, and if you develop a raised temperature. Both CVS and amniocentesis have a small increased risk of miscarriage above the normal risk of miscarriage at this gestation—around 1 percent each—with a slight increased risk for a CVS if performed earlier than 10 weeks and amniocentesis performed earlier

than 15 weeks. Some figures show that in general the miscarriage risk with CVS is slightly higher than it is with amniocentesis, particularly if your baby is small for his gestational age. Most miscarriages relating to these procedures occur within 72 hours of the test, but may occur up to two weeks later. Some people worry that those babies whose mothers have undergone CVS are at risk of being born without some of their fingers or toes. This risk is believed to exist only when CVS is performed at less than nine weeks pregnant. Your doctor should make you aware of all the risks before you consent to tests, and it is up to you as to whether or not to proceed.

Q Is there a safer noninvasive alternative to CVS or amniocentesis?

Noninvasive prenatal testing. (NIPT) is a blood test that analyzes tiny amounts of fetal DNA in the mother's blood. It is a simple test and the sample is sent for analysis. Early indications are that NIPT is around 99 percent accurate in the identification of babies who have chromosomal abnormalities. In the US this test is mainly offered to women who are at high risk for having a baby with Down syndrome or other abnormalities. It's typically offered from 10 weeks. The results take up to two weeks to come back. However, it's worth knowing that present statistics reveal that up to 5 percent of blood samples won't contain enough of the baby's DNA for proper analysis.

Q What diagnostic tests are available? Can they detect all fetal abnormalities?

CVS, amniocentesis, and cordocentesis, are diagnostic tests that will confirm whether or not your baby has specific abnormalities. As these tests are invasive, it is vital to know which one is right for you and what they involve.

Performed between 11 and 15 weeks, CVS can detect some chromosomal abnormalities, genetic abnormalities (such as in cystic fibrosis), musculoskeletal disorders (such as muscular dystrophy), blood disorders (such as sickle-cell anaemia, hemophilia or

thalassemia) and nervous system disorders (such as Tay Sachs disease). It can't tell you if your baby has a neural tube defect such as spina bifida. This is usually diagnosed through your anatomy scan (ultrasound) (see p.100), and it can be detected through amniocentesis.

CVS, AMNIOCENTESIS, CORDOCENTESIS, OR NIPT

The table below lists the diagnostic procedures, the window of time in which they are performed during your pregnancy, and the general conditions that they are testing for. Before undergoing any of these procedures talk to your doctor about the pros and cons.

PROCEDURE	WHEN	TESTING FOR
CVS (Invasive)	11-15 weeks	Chromosomal and genetic conditions, blood disorders, nervous system and musculoskeletal disorders.
Amniocentesis (Invasive)	from 14 weeks	Chromosomal and genetic conditions, blood disorders, neural tube defects, and musculoskeletal disorders.
Cordocentesis (Invasive)	18-24 weeks	Chromosomal and genetic conditions, blood disorders, neural tube defects, and musculoskeletal disorders.
NIPT (Noninvasive)	from 10 weeks	Chromosomal and genetic conditions. CVS and amniocentesis may be offered after this test.

Allow yourself time to **think about** which test you will have, if any, since this is an **important decision** and you need to be **prepared** as much as possible for the result.



Did you know...

Doctors use the following generalized statistics as a measure of how age might affect your risk of having a baby with Down syndrome. The risks are:

Age 20 the risk is **1 : 1,500**

Age 30 the risk is **1 : 800**

Age 35 the risk is **1 : 270**

Age 40 the risk is **1 : 100**

Age **45** the risk is greater than **1 : 50**